Unit 1

1. Introduction to AI

Module 1.4

1.4 AI Technologies

Tutorial 1.4

1.4 AI Technologies

AI-1.4.1 K1 Recognize the different technologies used to implement AI.

AI Technologies

AI can be implemented using a wide range of technologies such as:

- **Fuzzy logic**
  - Its a form of logic that allows reasoning with degrees of truth rather than classic logic of binary true/false.
  - Fuzzy logic is useful when dealing with uncertainty, vagueness, or imprecise information
  - It is similar to humans in thinking and making decisions.
  - Fuzzy logic examples
    - Control Systems - Air conditioners, washing machines, and traffic lights
    - Medical Diagnosis - Handling symptoms that aren’t black-and-white
    - Natural Language Processing - Understanding vague or nuanced language
    - AI & Robotics - Making decisions in uncertain environments

AI Technologies Cont.

- **Search algorithms**
  - Search algorithms are techniques to find specific data within a larger dataset or to explore possible solutions in a problem space.
  - Real-World Applications
    - Search engines (like Google)
    - Route planning (GPS systems)
    - Game AI (finding best moves)
    - Cybersecurity (password cracking algorithms)

AI Technologies Cont.

- **Reasoning techniques**
  - Rule engines
    - A rule engine in AI automates decision-making by applying a set of predefined if-then rules to incoming data
    - Rule engine is like giving an AI a playbook written by experts, so it can make consistent decisions without needing to learn from scratch like machine learning models
  - Deductive classifiers
    - These AI systems use logical reasoning to categorize data based on predefined rules and domain knowledge, rather than learning from examples like machine learning models do
  - Case-based reasoning
    - Case-Based Reasoning (CBR) is an AI technique that solves new problems by adapting solutions from similar past cases
    - It does not rely on abstract rules or training data, it mimics human thinking by recalling previous experiences and tweaking them to fit the current situation
    - For Example (AI patient diagnosis):
      - Past cases with similar symptoms are retrieved and treatment plan from that case is reused
      - AI then revises treatment plan based on current patient's unique conditions
      - Then, it retains the new case for future reference
  - Procedural reasoning
    - Procedural Reasoning is a method where intelligent agents make decisions and take actions based on procedural knowledge (knowledge about how to do things)

AI Technologies Cont.

- Machine learning techniques
  - Neural networks
    - Neural network is a type of machine learning model which is inspired by the structure and function of the human brain
    - It’s made up of layers of interconnected nodes which are called neurons, that work together to recognize patterns, make decisions, and learn from data
  - Bayesian models
    - Bayesian models are statistical frameworks that use Bayes’ Theorem to update beliefs or predictions based on new data
    - They treat probability as a measure of belief or certainty, rather than just frequency, making them powerful tools for reasoning under uncertainty
    - Example (Imagine you're trying to diagnose a disease)
      - Prior knowledge: The disease is rare
      - Likelihood: A test result indicates that the disease is present
      - Posterior: You combine both to estimate the actual chance if the patient has the disease

AI Technologies Cont.

- Machine learning techniques
  - Decision trees
    - Decision Trees are a popular machine learning technique used for both classification and regression tasks
    - It breaks down complex problems into a series of yes/no questions and forming a tree-like structure which leads to a final decision or prediction
    - How Decision Trees Work
      - Start at the Root Node (Represents the entire dataset)
      - Split Based on Features (Ask questions like “Is temperature > 30°C?” to divide data)
      - Branch Out (Each answer leads to a new node or decision)
      - Repeat Splitting (Continue until reaching the the leaf node which is final outcome)
  - Random forest
    - Its a machine learning algorithm which builds a “forest” of decision trees and combines their outputs to improve accuracy and reduce overfitting
    - How Random Forest Works
      - Build Multiple Decision Trees (Each tree is trained on a random subset of the data)
      - Random Feature Selection (When splitting nodes, each tree considers only a random subset of features)
      - Make Predictions:
        - Classification (Each tree votes for a class; the majority wins)
        - Regression (The average of all tree predictions is taken)

AI Technologies Cont.

- Machine learning techniques
  - Linear regression
    - It is a powerful tool in statistics and machine learning
    - It is used to model the relationship between a dependent variable and one or more independent variables, drawing the best straight line through a scatterplot of data to make predictions or understand trends
  - Logistic regression
    - Logistic Regression is a supervised machine learning algorithm used for classification tasks, especially when the output is binary (like yes/no, true/false, 0/1)
    - Unlike linear regression, which predicts continuous values, logistic regression estimates the probability that a given input belongs to a particular class
  - Clustering algorithms
    - Clustering algorithms are unsupervised machine learning techniques used to group data points into clusters, where items in the same cluster are more similar to each other than to those in other clusters
    - Very useful when labeled data is missing and you want to discover hidden patterns or struct

AI Technologies Cont.

- Machine learning techniques
  - Genetic algorithms
    - Genetic Algorithms (GAs) are optimization techniques inspired by natural selection and genetics
    - It simulates the process of biological evolution to find optimal solutions to problems
    - It is helpful in solving complex problems where traditional methods struggle
  - Support vector machine (SVM)
    - It is a supervised learning algorithm which is used for both classification and regression tasks
    - SVM is most commonly applied to classification problems
- AI-based systems typically implement one or more of these technologies

THANK YOU!!!